

Introduction to OOPS Programming

Total Time: 3 Hours

Section A: Concept Application (Theoretical – Scenario Based)

1. You are building a **food delivery application**. At which stage would you prefer using a **compiler** over an **interpreter**, and why? Explain based on execution speed, error detection and deployment
2. You are asked to design a program that **calculates employee salary** with multiple conditions (bonus, tax, overtime). Which programming constructs would you combine to solve this efficiently?
3. While developing a **menu-driven console application**, why are **loops preferred over conditional repetition**? Explain which loop you would choose and why?
4. A student writes a program to calculate average marks using int variables and gets incorrect output. Identify the probable cause and explain how choosing the correct **data type** solves the issue.
5. You are debugging a program that **complies successfully but crashes at runtime**. Which category of error is this? Mention **two debugging practices** you would follow to resolve it.
6. You need to store **student records** with name, marks, and grade. Why would you choose a **structure over multiple arrays** in this scenario?

Section B: Practical Coding Tasks

1. Conditional Logic Implementation

Design a **"Study Mood Assistant"** that:

- Takes user input: hours studied today
- Displays motivation messages based on conditions
- Uses **if-else ladder or switch case**

2. Loops + Arrays

Create a program that:

- Accepts 7 **days of screen time data**
- Calculates and prints:
 - Total screen time
 - Average screen time
- Displays a warning if average exceeds healthy limit

3. Functions & Pointers

Write a program to:

- Swap two numbers **using a user-defined function**
- Implement swapping using **pointers**
- Explain why pass-by-reference is necessary here

4. File handling

Build a program that:

- Writes daily goals to a file
- Reads and displays them on program restart
- Uses correct file modes

Section C: Mini Capstone Project

Mini Project: Student Productivity Tracker

Objective

Build a **console-based Student Productivity Tracker** that logs daily study hours and generates a weekly report.

Your application must:

1. Menu-driven program
2. Log daily study hours
3. Generate weekly report
4. Store data using files